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6.1 Introduction

Measurement is the act or the result of a quantitative comparison between a given quantity of same kind chosen as a standard. A device or mechanism used for comparing the unknown quantity with the unit of measurement or a standard quantity is called a measuring instrument.

Instruments which measure electrical quantities like voltage, current, power, energy etc. are called electrical instruments. These instruments are generally named after the electrical quantity to be measured. The instruments which measure voltage, current, electrical power and energy are called voltmeter, ammeter, wattmeter and energy meter respectively.

6.2 Classification of electrical measuring instruments

The various electrical measuring instruments may broadly classified into two groups, namely

1. Absolute instruments
2. Secondary instruments

6.2.1 Absolute instruments

Those instruments which indicate the quantity to be measured in terms of the instruments constants and its deflection are known as absolute instruments. It requires no previous calibration or comparison.

Example: Tangent galvanometer

6.2.2 Secondary instruments

Those instruments in which the value of electrical quantity to be measured can be measured from the deflection of instruments only are called secondary instruments. These instruments are calibrated by comparison with an absolute instrument or another secondary instrument which has already been calibrated against an absolute instrument.

Example: Voltmeters, ammeters and watt meters.

Secondary instruments are classified as:

i. Indicating instruments
ii. Integrating instruments
iii. Recording instruments

i. Indicating instruments

An instrument which indicates the magnitude of the electrical quantity being measured at the time at which it is being measured is called as an indicating instrument. It has a pointer which moves

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